Suggested problems: 1-17 odd, 23-29 odd, 35, 37, 41, 43

Problem 1: The temperature, in °C, at a point (x, y, z) is given by $T(x, y, z) = x^2 + y^2 + z^2$. What do the level surfaces of T look like and what do they mean in terms of temperature?

The level surfaces are concentric spheres, centered at the origin. Points along a particular level surface all have the same temperature.

Problem 2: What do the level surfaces of the following functions look like?

(a) $f(x, y, z) = x^2 + y^2$

The level surfaces are of the form $x^2 + y^2 = c$, where c is a constant. If $c < 0$, the level surface is empty. If $c = 0$, the level surface consists of the z-axis. If $c > 0$, the level surface is a cylinder wrapping around the z-axis of radius $\sqrt{c}$.

(b) $g(x, y, z) = z - y$

The level surfaces are of the form $z - y = c$, where c is a constant. Rearranging gives

$$z = y + c.$$

This is a linear function, written in slopes-intercept form. The slopes in the x-direction is 0, the slope in the y-direction is 1, and the z-intercept is c.

Problem 3: A company makes three sizes of cardboard boxes: small, medium, and large. It costs \$2.50 to make a small box, \$4.00 for a medium box, and \$4.50 for a large box. The fixed costs to make any number of boxes is \$8000.

(a) Express the cost, C, of making x small boxes, y medium boxes, and z large boxes as a function $C = f(x, y, z)$.

$C = f(x, y, z) = 2.5x + 4y + 4.5z + 8000$

(b) Find $f(3000, 5000, 4000)$ and interpret it.

$f(3000, 5000, 4000) = 53500$. This means its cost \$53,500 to manufacture 3000 small boxes, 5000 medium boxes, and 4000 large boxes.

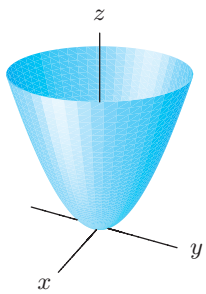

Figure 12.72: Elliptical paraboloid $z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

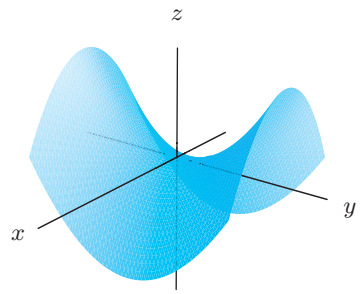

Figure 12.73: Hyperbolic paraboloid $z = -\frac{x^2}{a^2} + \frac{y^2}{b^2}$

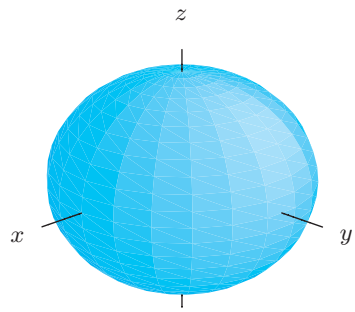

Figure 12.74: Ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

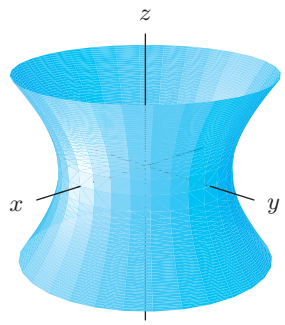

Figure 12.75: Hyperboloid of one sheet $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

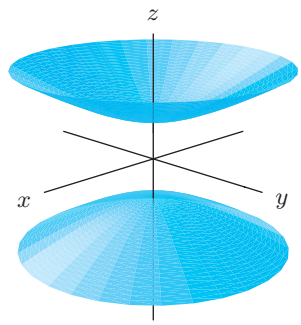

Figure 12.76: Hyperboloid of two sheets $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$

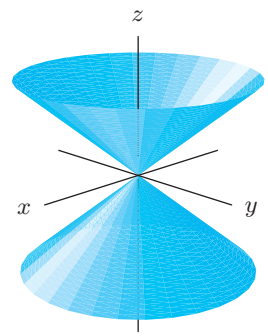

Figure 12.77: Cone $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$

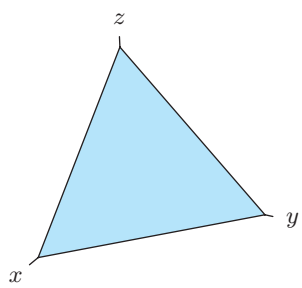

Figure 12.78: Plane $ax + by + cz = d$

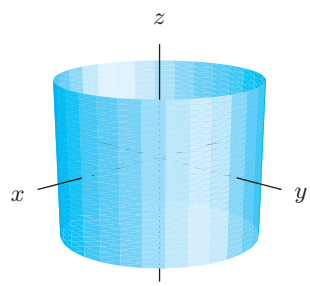

Figure 12.79: Cylindrical surface $x^2 + y^2 = a^2$

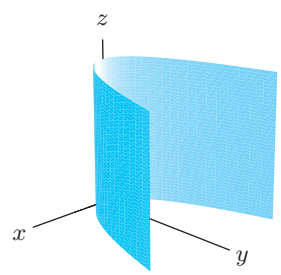

Figure 12.80: Parabolic cylinder $y = ax^2$

Problem 4: Use the figures above to identify the surfaces defined by the following equations.

(a) $z - 9x^2 - y^2 = 0$ elliptic paraboloid
(b) $z + x^2 - y^2 = 0$ hyperbolic paraboloid
(c) $-x^2 - 4y^2 + 1 = 0$ cylindrical surface
(d) $y - 2x^2 = 0$ parabolic cylinder

Problem 5: Describe the level surfaces of

$$f(x, y, z) = \frac{x^2}{4} + z^2$$

in words.

The level surfaces are of the form $\frac{x^2}{4} + z^2 = c$. If $c < 0$, the level surface is empty. If $c = 0$, the level surface is the y-axis. If $c > 0$, the level surface is a cylindrical surface wrapping around the y-axis.

Problem 6: Let

$$f(x, y, z) = x^2 - y^2 + z^2.$$

Describe the level surface $f(x, y, z) = 1$ in words.

The level surface is a hyperboloid of one sheet, wrapping around the y-axis.

Problem 7: Find a function $g(x, y, z)$ whose level surface with $g = 0$ is the graph of $f(x, y) = x + 2y$. Now find a different function $h(x, y, z)$ whose level surface with $h = 1$ is the graph of $f(x, y) = x + 2y$.

$g(x, y, z) = x + 2y - z$ and $h(x, y, z) = x + 2y - z + 1$

Problem 8: Define $g(x, y, z) = x^2 + y^2 + z^2$. Can the level surface defined by $g = 10$ be written as the graph of a single function $f(x, y)$?

No. For if this were true, we would need to have $z = f(x, y)$. But solving $x^2 + y^2 + z^2 = 10$ for z gives *two* values of z when we take the square root. Thus a single function f does not exist.